



MASENO UNIVERSITY
UNIVERSITY EXAMINATIONS 2024/2025

**THIRD YEAR FIRST SEMESTER EXAMINATIONS FOR THE
DEGREE OF BACHELOR OF SCIENCE IN HORTICULTURE,
BACHELOR OF SCIENCE IN AGRONOMY, BACHELOR OF
SCIENCE IN AGRICULTURAL EDUCATION AND BACHELOR OF
SPECIAL NEEDS EDUCATION WITH INFORMATION
TECHNOLOGY**

MAIN CAMPUS

AHC 301: FARM POWER AND MACHINERY

Date: 13th January, 2025

Time: 8.30 - 11.30 am

INSTRUCTIONS:

- DO NOT write anywhere on this question paper
- Answer ALL questions in SECTION A and any other TWO from SECTION B



2024/2025 ACADEMIC YEAR

THIRD YEAR, FIRST SEMESTER EXAMINATIONS FOR THE DEGREE OF
BACHELOR OF SCIENCE IN HORTICULTURE, BACHELOR OF SCIENCE IN
AGRONOMY, BACHELOR OF SCIENCE IN AGRICULTURAL EDUCATION AND
Bed (SCIENCE, SPECIAL NEEDS EDUCATION)

AHC 301: FARM POWER AND MACHINERY

TIME: 3 HOURS

INSTRUCTIONS:

The paper is composed of two sections (A & B)

Answer ALL question in section A and TWO questions in section B

SECTION A (40 Marks): Short Answer questions. Answer ALL the questions in this section

1. True/False: In a four-stroke petrol engine, the power stroke occurs every two revolutions of the crankshaft. (2 Mark)
2. True/False: In a two-stroke engine, the intake and exhaust processes occur simultaneously during part of the power cycle. (2 Mark)
3. True/False: The compression ratio in a diesel engine is typically higher than in a petrol engine. (2 Mark)
4. True or False: The crankshaft is responsible for converting the reciprocating motion of the pistons into rotational motion. (2 Marks)
5. True or False: The camshaft in a tractor engine controls the opening and closing of the exhaust and intake valves. (2 Marks)
6. True or False: The primary purpose of primary tillage is to bury weed seeds and reduce weed pressure in the field. (2 Marks)
7. What is a general purpose tractor and how are they different from row crop tractors? (2 Marks)
8. Describe how diesel engines ignite the air-fuel mixture and contrast this with the ignition process in petrol engines. (2 Marks)
9. Describe the lubrication process in a two-stroke engine and explain why oil must be mixed with the fuel. What are the environmental implications of this lubrication method? (4 Marks)
10. What are the primary functions of a moldboard plow in tillage operations, and how does its design contribute to soil turnover and weed control? (4 Marks)
11. Describe the correct sequence of strokes in a four-stroke cycle and explain the function of each stroke in the engine's operation. (4 Marks)

12. Discuss the two advantages and disadvantages of using renewable energy sources, such as solar and wind power, in agricultural applications. (4 Marks)
13. Briefly discuss what you understand by "Farm Mechanization" and five challenges facing its adoption in Kenya. (8 Marks)

SECTION B (30 MARKS): Answer TWO questions from this section

- 14.
- a) Explain the role of tractors as a power source in agriculture and describe some of the key tasks they perform on a farm. (5 Marks)
 - b) Suggest and explain five solutions to address challenges of Agricultural Mechanization in Kenya. (10 Marks)
- 15.
- a) Explain how a chisel plow differs from a moldboard plow in terms of soil disturbance and residue management. (5 Marks)
 - b) Describe the operational role of a disc plow in primary tillage. In what types of soil conditions is it most effective, and what are its advantages over other plows? (5 Marks)
 - c) How does a subsoiler work to alleviate soil compaction, and what impact does it have on root development and water infiltration in crop production? (5 Marks)
- 16.
- a) Explain the role of threshers in post-harvest processing. How does threshing improve the efficiency of crop handling and storage? (5 Marks)
 - b) Describe the purpose of a combine harvester in post-harvest operations. How does it integrate multiple functions, and what are the benefits of this integration? (5 Marks)
 - c) How do sorting and grading machines enhance the quality and marketability of harvested produce? Provide examples of how these machines work in post-harvest processing. (5 Marks)